D03 · Yield & Loss Reduction

04 · Value stream mapping & current-future state

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1 WHAT IT IS

Value stream mapping (VSM) is the structured technique for visualizing the end-to-end material and information flow that delivers a product family to the customer — typically drawn as a current-state map, then redesigned as a future-state map, with a transition plan that converts one into the other. The technique was codified in the seminal text on the subject and remains the most diagnostic flow-design tool available; its utility depends entirely on whether the future state is implemented.

2 WHY IT MATTERS

VSM is the most powerful flow-design tool in the Lean kit and the one most consistently reduced to ritual. **The dominant failure mode is the VSM workshop run as a standalone event — current state mapped, future state aspired, transition plan drafted, then nothing happens because no one owns the transition**. The reference text on VSM treats this directly **(Rother & Shook, 2003 — peer-reviewed, "Learning to See", Lean Enterprise Institute)**: the mapping is the easy part; the implementation of the future state requires a value-stream manager with cross-functional authority, a sequenced transition plan with named owners and dates, and a sustained execution cadence. Plants that produce maps without that infrastructure capture roughly **10–15% of the gap between current and future state**; plants that produce maps with a value-stream manager and disciplined execution capture **60–80%**. The asymmetry is enormous, and the difference is structural, not methodological.

3 HOW IT IS DONE

1. **Select the product family before drawing — a VSM that tries to cover the entire portfolio is a VSM that will not be implementable.** A product family is a group of products that move through similar processing steps in similar order; typical plants have 3–8 families. The first VSM should be the highest-revenue family or the family with the largest cost-of-quality gap, not the family that is easiest to map. Family selection determines whether the map produces an actionable transformation or an interesting wall poster.
2. **Map current state by walking the value stream from customer back to supplier — not by interviewing the engineering team.** The current-state map captures what actually happens, not what is supposed to happen. Walk the flow in reverse (from finished-goods shipping back to raw-material receiving), with a stopwatch and a count of WIP at every queue, capturing real lead times, real cycle times, real changeover times, real defect rates. Engineering's understanding of the flow and the floor's reality usually diverge by 30–60%; the map must capture reality.
3. **Design the future state with takt time as the anchor, then work backward to flow design.** Takt time (available production time divided by customer demand rate) is the heartbeat the value stream must match. Once takt is set, the future-state design decisions follow: where can true continuous flow happen (and where not), where do FIFO lanes belong, where is the pacemaker process, where does the pull signal originate. Future-state design without takt as the anchor produces aspirational maps disconnected from operational reality.
4. **Assign a value-stream manager with cross-functional authority — not a coordinator with a tracker.** The value-stream manager owns the transition from current to future state across the functional silos (production, supply chain, quality, engineering, finance) that the value stream crosses. The role requires actual authority to make trade-offs and decisions; an "owner" who needs to escalate every decision is not an owner. Without the role, the transition stalls at the first functional disagreement.
5. **Break the transition into 60–90 day kaizen bursts — not a 24-month transformation plan.** A future state that is 18 months out has no force on operational decisions today; a sequenced series of 60–90 day kaizens, each implementing a piece of the future state, builds momentum and verifies the design assumptions incrementally. Each kaizen produces a measurable shift in lead time, WIP, or FPY that validates the trajectory before the next kaizen launches.
6. **Re-map every 12–18 months — value streams drift, and the future-state map of last year is the current-state map of next year.** A VSM is not a permanent artifact; it is a snapshot of a moving system. Demand patterns shift, products evolve, suppliers change, capacity invests. The value stream that was 80% redesigned last year is now 20% out of alignment with new demand reality. The discipline of periodic re-mapping is what prevents the cycle of one heroic transformation followed by

years of degradation.

WORKED EXAMPLE — ILLUSTRATIVE

An electronics contract manufacturer (1 plant, 14 product families) runs a 3-day VSM workshop on its highest-revenue family. Current state mapped: **lead time 28 days, value-add time 11 hours (1.6% of total), 9 process steps with WIP queues averaging 3.2 days each**. Future state designed: **9-day lead time, 5 process steps consolidated, 2 pull loops, 1 supermarket**. The workshop ends with a 24-month transition plan and an assigned "VSM champion" (a quality engineer with no operational authority). Eighteen months later, lead time is still 28 days; the future state map is on a wall but nothing has been implemented. The re-launch installs a value-stream manager (the operations manager for that family, reporting directly to the plant manager), breaks the transition into **6 sequenced 60-90 day kaizens**, and assigns finance to validate the lead-time and WIP outcomes monthly. Results 9 months after the re-launch: **lead time 28 -> 11 days (-61%), WIP $7.4M -> $3.1M (-58%), on-time delivery 84% -> 96%, the future state is 78% implemented**. The first 18 months produced a map; the next 9 months produced an operation. Same map, different infrastructure.

IN PRACTICE

Value-stream manager with cross-functional authority and direct senior reporting

Direct reporting to the plant manager (or COO in multi-plant operations), explicit authority over trade-offs within the value stream, a P&L view of the value stream's economic performance **(Rother & Shook, 2003 — peer-reviewed, Lean Enterprise Institute)**. Without that, the role is a coordinator who can convene meetings but cannot make decisions.

Walk current state customer-to-supplier, not engineering-to-engineering

Map from the customer arrival backward, on the floor, with the operator's perspective — not from the engineering team's conceptual understanding. The flow that engineering thinks exists is rarely the flow that operates.

Design future state with takt as the anchor, not capacity

Future state is designed against customer takt — the rate at which customers demand the product — not against the capacity of the most expensive machine. Designing to capacity produces the current state polished; designing to takt produces flow.

Re-map every 12–18 months; the value stream drifts

Mix changes, suppliers change, customer cadence shifts. A future state implemented in Q1 is the current state by Q3 of the following year. Plants that map once and treat it as done are plants that miss the next bottleneck migration.

WHAT YOU WOULD USE

VSM & TRANSITION MANAGEMENT

iGrafx Process / Sparx Enterprise Architect — Enterprise process-mapping platforms with native VSM iconography (data boxes, push arrows, supermarkets, FIFO lanes), suitable for organizations standardizing flow design across multiple plants and value streams **(iGrafx — vendor)**.

Lucidchart / Miro / Visio — General-purpose diagramming tools widely adopted for VSM because of their collaboration features; the right fit when the value-stream work is being done by an in-house team that can apply the VSM convention without specialized software **(Lucid — vendor; Microsoft — vendor)**.

KaiNexus / LeanKit / Mevisio — Lean management platforms with VSM modules paired with kaizen-pipeline tracking and value-stream KPI dashboards — the right fit for the transition management piece, not the mapping itself **(KaiNexus — vendor; Mevisio — vendor)**.

Brown paper, sticky notes, and a disciplined value-stream walk — The original VSM technique, used by Toyota and codified by Rother & Shook. The lowest-tech option produces the highest team engagement; the artifact is photographed and digitized for follow-on tracking, but the mapping itself happens in the gemba with paper.

4 SO WHAT

The value stream map without an assigned manager and a sequenced transition pipeline is a map that ends in a folder; the operation remains the operation. Select the product family before mapping, walk the current state from customer to supplier not from engineering's understanding, design the future state with takt as the anchor, assign a value-stream manager with real cross-functional authority, break the transition into 60–90 day kaizen bursts, and re-map every 12–18 months as the value stream drifts. The mapping is intellectually engaging; the manager is what makes the map implementable. The 5–6x difference between plants that capture the gap and plants that produce wall posters lives entirely in that structural choice.

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